Lecture 8 - Continuity and Differentiability

Garrett Credi

10/01/2026

Announcements

1. Exam 1 is next Tuesday (10/06/2026) in class.
2. Exam 1 Review Session is next Monday (10/05/2026) from 6:10-8:10 PM in Vincent Hall 1.
3. All quizzes will be graded by Friday Night. If you want me to look at your homeworks, please email me and I'll grade them specifically.
4. Practice Exam Solutions will be released after class today.
5. Compiled lecture slides also released after class today.

Quiz Time

Quiz Time! Let's get into neat rows and spread out - only then will I pass out the quizzes. DO NOT START OR FLIP OVER THE QUIZ UNTIL I OFFICIALLY START THE QUIZ. If you have any questions about the rules for quizzes, refer to the slide in Lecture 2 or ask me directly. No cheating as per usual.

Continuity

Continuity Recap I

Recall our definitions of continuity for a function $f(x)$:

1. $f(x)$ is continuous at $x = a$ if:

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Continuity Recap II

Also recall that: if $f(x)$ and $g(x)$ are continuous at $x = a$ and $h(x)$ is continuous at $f(a)$ then:

- $f(x) + g(x)$ is continuous at $x = a$
- $f(x) - g(x)$ is continuous at $x = a$
- $f(x) \cdot g(x)$ is continuous at $x = a$
- If $g(a) \neq 0$, then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$
- $h(f(x))$ is continuous at $x = a$.

Continuity Recap III

Finally we determined a large family of continuous functions:

- Constant functions $f_c(x) = c$
- Identity function $f(x) = x$
- Polynomials
  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$
- Root functions $f(x) = \sqrt[n]{x}$
- Rational functions $\dfrac{p(x)}{q(x)}$
- Trigonometric functions
  $\sin(x), \cos(x), \tan(x), \arcsin(x)$ etc.
- Exponential functions $f(x) = c^x$
- Logarithmic functions $f(x) = \log_c(x)$

Classifying Failures of Continuity: Jump Discontinuities

What does discontinuity look like?

Jump Discontinuity

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3. Those values do not agree:

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2. The function limits towards a value from the right: $\lim_{x \to a^+} f(x)$ exists
3. Those values do not agree:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x) \tag{1}$$

Classifying Failures of Continuity: Jump Discontinuities Visually

Visually, jump discontinuities look like:

Classifying Failures of Continuity: Removable Discontinuities

Best kind of discontinuity.

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2. $f(a)$ either does not exist

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A function $f(x)$ has a *removable discontinuity* at $x = a$ if:

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2. $f(a)$ either does not exist (we specifically call this a *hole discontinuity* or just a *hole*) or is not equal to $\lim_{x \to a} f(x)$.

Can redefine $f(x)$ *at a single point* to make it continuous at $x = a$.

Removable Discontinuities Visually

Visually,

Classifying Failures of Continuity: Infinite Discontinuities

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A function $f(x)$ has an *infinite discontinuity* if, equivalently:

- $f(x)$ has a vertical asymptote at $x = a$ or
- $\lim_{x \to a} |f(x)| = +\infty$

Infinite Discontinuities Visually

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Classifying Failures of Continuity: Essential Discontinuities

The *WORST* kinds of discontinuities.

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- At least one of the one-sided limits do not exist (not even diverging to $\pm infty$) or
- As $x \to a$, $f(x)$ takes on every number in some interval (v_{min}, v_{max}) *infinitely many times*

Essential Discontinuities Visually

Visually,

What does Continuity buy us?

We already saw one important use of continuous functions:

Direct Substitution

Intermediate Value Theorem

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If $f(x)$ is continuous at $x = a$ then:

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Continuous functions also have another very important property:

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Tagline: all intermediate output values must be hit by intermediate input values.

IVT Example: Root finding I

Question: For the function

$$f(x) = x^3 - 3x^2 + 2x + 1, \tag{3}$$

which of the following intervals must contain a root of $f(x)$?

1. $[-2, -1]$
2. $[-1, 0]$
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5. $f(2) = 8 - 12 + 4 + 1 = 1$

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4. $f(1) = 1 - 3 + 2 + 1 = 1$
5. $f(2) = 8 - 12 + 4 + 1 = 1$

Only know that a root must be inside $[-1, 0]$.

Differentiability

Differentiability Recap

Again, recall that:

Differentiability Recap

Again, recall that:

1. $f(x)$ is differentiable at $x = a$:
2. $f(x)$ is differentiable on (a, b):
3. $f(x)$ is differentiable everywhere:
4. $f(x)$ is differentiable

Classifying Failures of Differentiability: Discontinuities

We can already classify some behavior making functions differentiable.

Classifying Failures of Differentiability: Discontinuities

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Recall: if $f(x)$ is not *continuous* at $x = a$ then it cannot be *differentiable* at $x = a$.

Classifying Failures of Differentiability: Corners

We can classify the rest of the behavior by examining the *continuity* of the derivative function $f'(x)$:

Corner Singularity

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For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *corner singularity* or *corner* at $x = a$

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For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *corner singularity* or *corner* at $x = a$ if the derivative function $f'(x)$ has a *jump discontinuity* at $x = a$.

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Visually,

Classifying Failures of Differentiability: Cusps

Similarly,

Cuspidal Singularity

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For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *cuspidal singularity* or *cusp* at $x = a$ if the derivative function $f'(x)$ has an *infinite discontinuity* at $x = a$

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Classifying Failures of Differentiability: Vertical Tangents

Vertical Tangent Singularity

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *vertical tangent singularity* or a *vertical tangent* at $x = a$

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$$\lim_{x \to a} |f'(x)| = +\infty. \tag{5}$$

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Classifying Failures of Differentiability: Endpoint Singularity

Kind of a non-problem singularity:

Endpoint Singularity

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For a function $f(x)$ defined on a *closed* interval $[a, b]$,

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For a function $f(x)$ defined on a *closed* interval $[a, b]$, we say that $f(x)$ has an *endpoint singularity* at the endpoints $x = a$ and $x = b$

Classifying Failures of Differentiability: Endpoint Singularity

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Classifying Failures of Differentiability: Endpoint Singularity

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Intuituion: Functions whose graphs are continuous, and corner/cusp/vertical tangent free.

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Mean Value Theorem

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- $$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(a). \tag{6}$$

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3. Lots of the really useful tools in multivariable calculus need to use MVT to even work.

“Fun” application:

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks!